

VIKAS SHARMA

Python Developer | Backend Engineering | Open Source (PyPI) | Distributed Systems

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PROFESSIONAL SUMMARY

Python Developer with 3+ years building production-grade backend systems, SDKs, and data platforms. Author of **pyzdata** on PyPI (Production/Stable, Python 3.10-3.13, PEP 561 typed, 90 unit tests). Architected distributed systems processing **\$48M+ in daily transaction throughput across 150+ connected clients** and delivered **150x performance speedups** via NumPy vectorization and parallel orchestration. Strong in Python internals, async/concurrent systems, API design, and financial data engineering.

TECHNICAL SKILLS

Languages: Python (Expert; object-oriented design, type hints), C++, SQL, Bash

Python Ecosystem: NumPy, Pandas, Polars, DuckDB, Numba, Dask, AsyncIO, Multiprocessing, Multithreading, pytest, Pydantic, Click

Web & APIs: Django, Flask, FastAPI, Streamlit, REST APIs, WebSocket, Selenium

Data & Storage: Parquet, DuckDB, SQL, caching strategies, ETL pipelines, vectorized computation

Systems & Infra: ZeroMQ, microservices, message-passing, FileLock orchestration, Docker, AWS EC2, Linux, Git

Packaging & CI/CD: PyPI publishing, semantic versioning, GitHub Actions, trusted publishing (Sigstore), PEP 561 typing

PROFESSIONAL EXPERIENCE

Software Engineer II - Pranjali Growcap Pvt. Ltd.

Feb 2024 - Present

Proprietary fintech firm - high-throughput transaction processing

New Delhi, India

- **Python SDK Architecture:** Designed a Python SDK powering **150+ connected clients with \$48M in daily transaction throughput** - integrating 3 external REST/WebSocket APIs with real-time validation, reconciliation, P&L computation, and audit trails. Eliminated 100% of manual workflows.
- **Distributed Systems:** Engineered a **ZeroMQ hub-spoke architecture** - central message broker coordinating 5 independent Python worker processes; CSV-backed state machines for crash recovery, smart request routing with retry and stale-response detection; fully autonomous on AWS EC2.
- **Performance Engineering (150x speedup):** Rebuilt a data-intensive batch processing engine with NumPy vectorization, $O(\log n)$ binary-search lookups, and multi-process parallel orchestration - cut runtime from **~150 hours to under 1 hour** across 16 worker processes coordinated via FileLock.
- **Parallel ETL & Analytics:** Processed **50,000+ parameter combinations x 250+ dates** on 8+ years of tick-level financial time-series data; built Polars + DuckDB aggregation layer with interactive Streamlit dashboards and automated Excel exports via xlwings.

Software Engineer - Pranjali Growcap Pvt. Ltd.

Jul 2022 - Jan 2024

Proprietary fintech firm

New Delhi, India

- Built v1 of the firm's Python platform SDK integrating multiple third-party REST and WebSocket APIs with real-time data streaming, session management, and automated retry logic.
- Developed in-house data-analysis framework processing tick-level financial datasets, replacing ad-hoc spreadsheet workflows with a versioned, reproducible pipeline.
- Shipped internal monitoring stack using Streamlit + Telegram bots for real-time alerts, dashboards, and system-health checks used daily by internal teams.
- Designed ETL pipelines for cleaning and structuring high-volume tick-level market data from multiple upstream feeds.

PROJECTS

PyZData (pip install pyzdata) - Published on PyPI | *v1.0.6, Production/Stable, MIT License*

- Python library + CLI + Streamlit web app for historical and intraday financial data. Parallel downloads, disk caching, rate-limiting, typed exception hierarchy, Config-driven environment. **32 GitHub stars, 15 forks, 90 unit tests, PEP 561 typed, trusted publishing via GitHub Actions (Sigstore).**

KiteWeb (28 stars, 9 forks) - Unofficial Python client for Zerodha Kite web endpoints: authentication, session management, and order and market-data calls; adopted across the Indian retail algorithmic-trading community.

Developer Productivity Tools - Jupyter context-menu launcher (Windows shell integration), PickleParquetReader (.pkl/.parquet to Excel), TerminalSequence (multi-monitor window arranger), Tele_Easy_Bot (Telegram wrapper).

EDUCATION

BSc Computer Science - Shaheed Sukhdev College of Business Studies, University of Delhi

2020 - 2023

CODING PROFILES

Data structures and algorithms practice: **CodeChef:** 2-star (rating 1495, peak 1559), 252 problems, 10 contests **LeetCode:** 65 solved (43 easy, 21 medium, 1 hard) **HackerRank:** SQL Intermediate + 3 verified certs